

# CSE 4417 IoT Theory

## Lecture 02: Applications of IoT

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bjedchaj



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# Alternate Definition

- The Internet of Things (IoT) is the network of physical objects that contain embedded technology to communicate and sense or interact with their internal states or the external environment

# IoT Applications

## **Business/Manufacturing**

Real-time analytics of supply chains and equipment, robotic machinery,

## **Healthcare**

Portable health monitoring, electronic recordkeeping, pharmaceutical safeguards.

## **Retail**

Inventory tracking, smartphone purchasing, anonymous analytics of consumer choices.

## **Security**

Biometric and facial recognition locks, remote sensors.

# IoT Applications

## **ATM**

These ubiquitous money dispensers went online for the first time way back in 1974.

## **WEB**

World Wide Web made its debut in 1991 to revolutionize computing and communications.

## **SMART METERS**

The first power meters to communicate remotely with the grid were installed in the early 2000s.

## **DIGITAL LOCKS**

Smartphones can be used to lock and unlock doors remotely, and business owners can change key codes rapidly to grant or restrict access to employees and guests.

# IoT Applications

- Smart Parking
- Structural health
- Noise Urban Maps
- Smartphone Detection
- Traffic Congestion
- Smart Lighting
- Waste Management
- Smart Roads
- River Floods
- Smart Grid
- Tank level
- Photovoltaic Installations
- Water Flow
- Silos Stock Calculation
- Perimeter Access Control
- Liquid Presence

# IoT Applications

- Forest Fire Detection
- Air Pollution
- Snow Level Monitoring
- Landslide and Avalanche Prevention
- Earthquake Early Detection
- Water Leakages
- Radiation Levels
- Explosive and Hazardous Gases
- Supply Chain Control
- NFC Payment
- Intelligent Shopping Applications
- Smart Product Management

# IoT Enabler - Connectivity

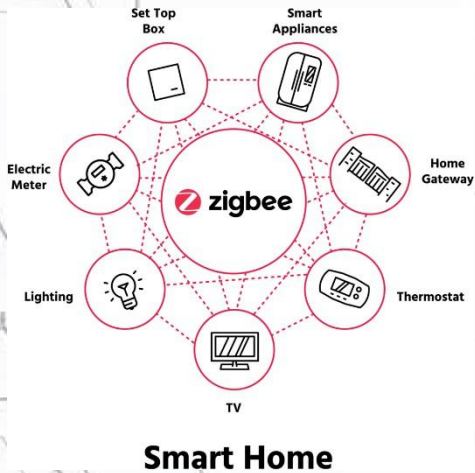
A number of technologies that are used for connection -

- RFID
  - Radio-frequency identification (RFID) uses electromagnetic fields to automatically identify and track tags attached to objects. An RFID system consists of a tiny radio transponder called a tag, a radio receiver, and a transmitter. When triggered by an electromagnetic interrogation pulse from a nearby RFID reader device, the tag transmits digital data, usually an identifying inventory number, back to the reader. This number can be used to track inventory goods. Passive tags are powered by energy from the RFID reader's interrogating radio waves. Active tags are powered by a battery and thus can be read at a greater range from the RFID reader, up to hundreds of meters.



# IoT Enabler - Connectivity

- Zigbee
  - Zigbee is an IEEE 802.15.4-based specification for a suite of high-level communication protocols used to create personal area networks with small, low-power digital radios, such as for home automation, medical device data collection, and other low-power low-bandwidth needs, designed for small scale projects which need wireless connection. Hence, Zigbee is a low-power, low-data-rate, and close proximity (i.e., personal area) wireless ad hoc network.





# IoT Enabler - Connectivity

- Bluetooth



- Bluetooth is a short-range wireless technology standard that is used for exchanging data between fixed and mobile devices over short distances and building personal area networks (PANs). In the most widely used mode, transmission power is limited to 2.5 milliwatts, giving it a very short range of up to 10 metres (33 ft). It employs UHF radio waves in the ISM bands, from 2.402 GHz to 2.48 GHz.[3] It is mainly used as an alternative to wired connections to exchange files between nearby portable devices and connect cell phones and music players with wireless headphones.

# IoT Enabler - Connectivity

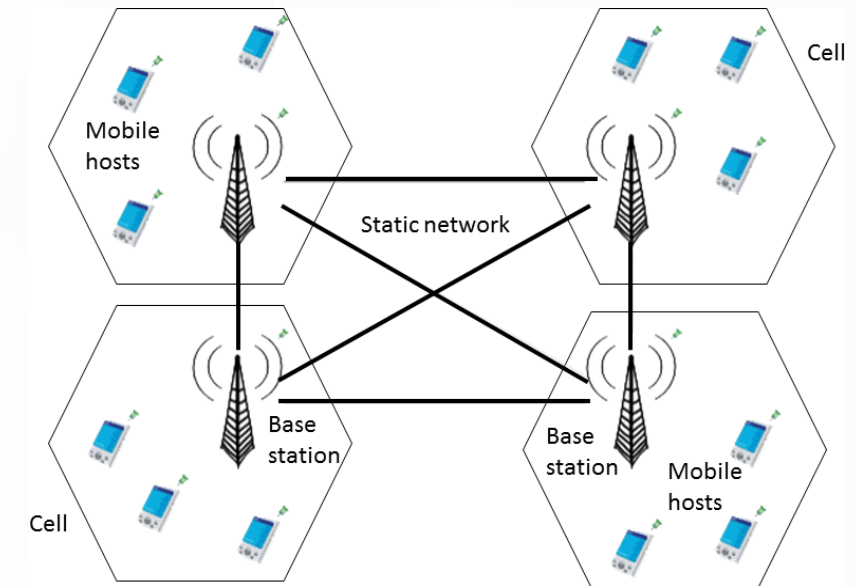
- WiFi

- Wi-Fi is a family of wireless network protocols based on the IEEE 802.11 family of standards, which are commonly used for local area networking of devices and Internet access, allowing nearby digital devices to exchange data by radio waves. These are the most widely used computer networks, used globally in home and small office networks to link devices and to provide Internet access with wireless routers and wireless access points in public places such as coffee shops, hotels, libraries, and airports to provide visitors.



# IoT Enabler - Connectivity

- Cellular network
  - A cellular network or mobile network is a telecommunications network where the link to and from end nodes is wireless and the network is distributed over land areas called cells, each served by at least one fixed-location transceiver (typically three cell sites or base transceiver stations). These base stations provide the cell with the network coverage which can be used for transmission of voice, data, and other types of content.





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# Thank You

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